

Departement Elektriese, Elektroniese en
Rekenaar-Ingenieurswese

Department of Electrical, Electronic and
Computer Engineering

Eksamen

Kopiereg voorbehou

Examination

Copyright reserved

Module EBN122
Elektrisiteit en Elektronika
18 November 2013

Module EBN122
Electricity and Electronics
18 November 2013



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Eksamensinligting:

Exam information:

Maksimum punte: <i>Maximum marks:</i>	90	Volpunte: <i>Full marks:</i>	90
Duur van vraestel: <i>Duration of paper:</i>	180 minute <i>180 minutes</i>	Oopboek / toeboek: <i>Open / closed book:</i>	Toe <i>Closed</i>
Totale aantal bladsye (hierdie blad ingesluit): <i>Total number of pages (including this page):</i>		26	

BELANGRIK- IMPORTANT

- Die eksamenregulasies van die Universiteit van Pretoria geld.
The examination regulations of the University of Pretoria apply.
- Alle vrae moet op die ANTWOORDBLAD beantwoord word.
All questions must be answered on the ANSWER SHEET.
- Studente mag hul eie rowwe berekeninge op die vraestel doen – die vraestel word nie ingeneem nie.
Students may do their own rough calculations on the question paper – the question paper will not be taken in.
- Antwoorde moet eenhede insluit!
Answers must include units!
- Polêre vorm: $A\angle\theta$; Reghoekige vorm: $a + jb$
Polar form: $A\angle\theta$; Rectangular form: $a + jb$
- Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie.**
Final answers must be written as real numbers, and not as fractions.

Dosent(e): <i>Lecturer(s):</i>	J. Joubert
	J.D. le Roux
Eksterne eksaminator: <i>External examiner:</i>	J.W. Odendaal

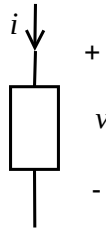
VRAAG/QUESTION 1 [18]

Vraag 1.1 [6]

- a) Die spanning en stroom by die terminale van die stroombaan is nul vir $t < 0$. Vir $t \geq 0$ is dit $v = 2e^{-2t}$ V en $i = 10 - 2e^{-2t}$ A. Bepaal die drywing by $t = 0.2$ s.
- b) Die spanning en stroom by die terminale van die stroombaan is nul vir $t < 0$. Vir $t \geq 0$ is dit $v = 2 + t$ V en $i = t - 1$ A. Hoeveel energie word aan die stroombaan gelewer tussen 0 en 3s.
- c) 'n 1.2 kW broodrooster neem 5 min om 4 snye brood te rooster. Bepaal die koste om 100 snye brood te rooster as energie 50 sent/kWh kos.

Question 1.1 [6]

- a) The voltage and current at the terminals of the circuit are zero for $t < 0$. For $t \geq 0$ they are $v = 2e^{-2t}$ V and $i = 10 - 2e^{-2t}$ A. Find the power at $t = 0.2$ s.
- b) The voltage and current at the terminals of the circuit are zero for $t < 0$. For $t \geq 0$ they are $v = 2 + t$ V and $i = t - 1$ A. Determine the energy delivered to the circuit between 0 and 3s.
- c) A 1.2 kW toaster takes 5 minutes to toast 4 slices of bread. Determine the cost to toast a 100 slices of bread if the cost of energy is 50 cents/kWh.

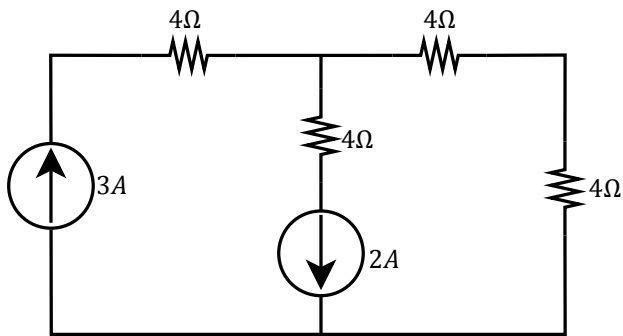


Vraag 1.2 [4]

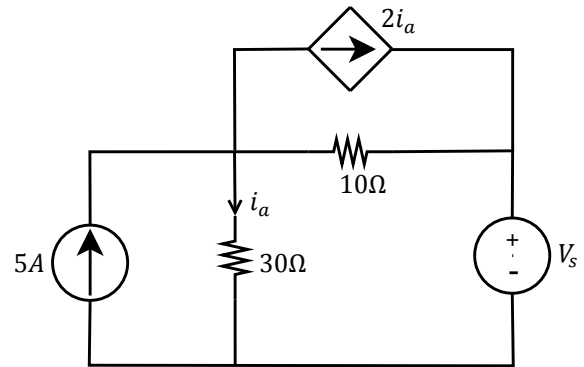
- a) Bereken die drywing geassosieer met die 3A en 2A stroombronne.
- b) Indien $i_a = 2A$, bereken die drywing geassosieer met die 5A stroombron, en die spanningsbron V_s .

Question 1.2 [4]

- a) Determine the power associated with the 3A and 2A current sources.
- b) If $i_a = 2A$, determine the power associated with the 5A current source and the voltage source V_s .



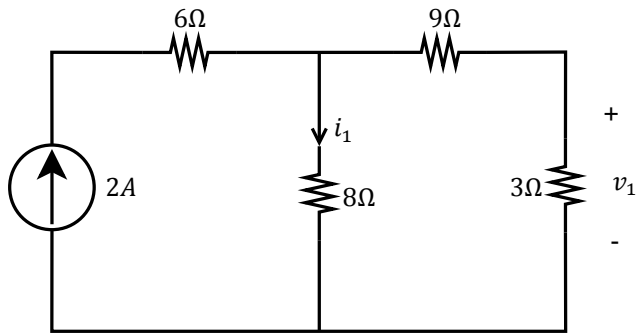
(a)



(b)

Vraag 1.3 [4]

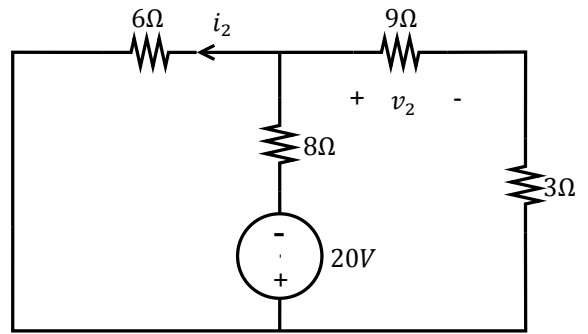
- a) Bereken v_1 en i_1 .
b) Bereken v_2 en i_2 .



(a)

Question 1.3 [4]

- a) Determine v_1 and i_1 .
b) Determine v_2 and i_2 .



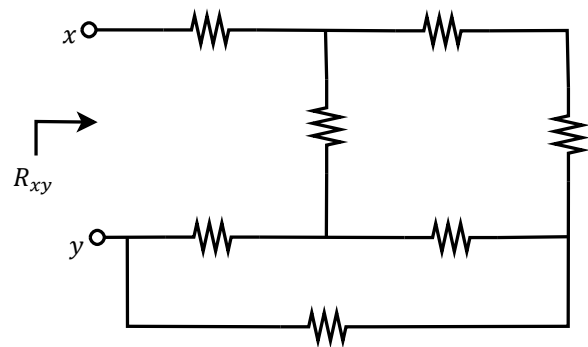
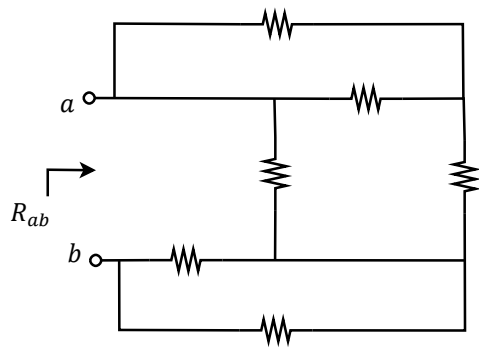
(b)

Vraag 1.4 [4]

Bereken R_{ab} en R_{xy} . Aanvaar alle weerstande het 'n waarde van 3Ω .

Question 1.4 [4]

Determine R_{ab} and R_{xy} . Assume all resistors have a value of 3Ω .



VRAAG/QUESTION 2 [32]

Vraag 2.1 [4]

Gebruik node-analise om twee vergelykings van die volgende vorm op te stel:

$$av_1 + bv_2 = -30 \text{ V} \quad (\text{node 1})$$

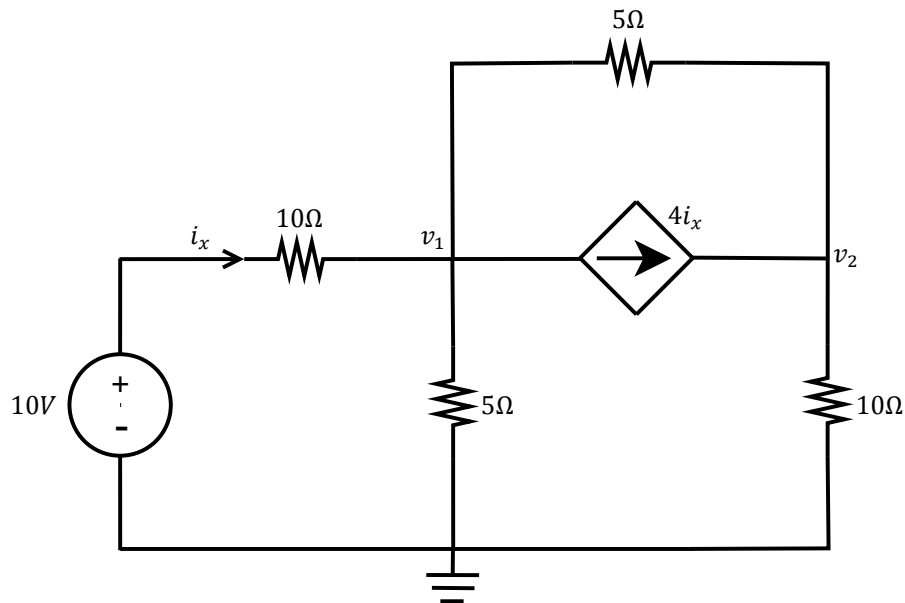
$$cv_1 + dv_2 = 40 \text{ V} \quad (\text{node 2})$$

Question 2.1 [4]

Use nodal analysis to set up two equations of the following form:

$$av_1 + bv_2 = -30 \text{ V} \quad (\text{node 1})$$

$$cv_1 + dv_2 = 40 \text{ V} \quad (\text{node 2})$$



Vraag 2.2 [4]

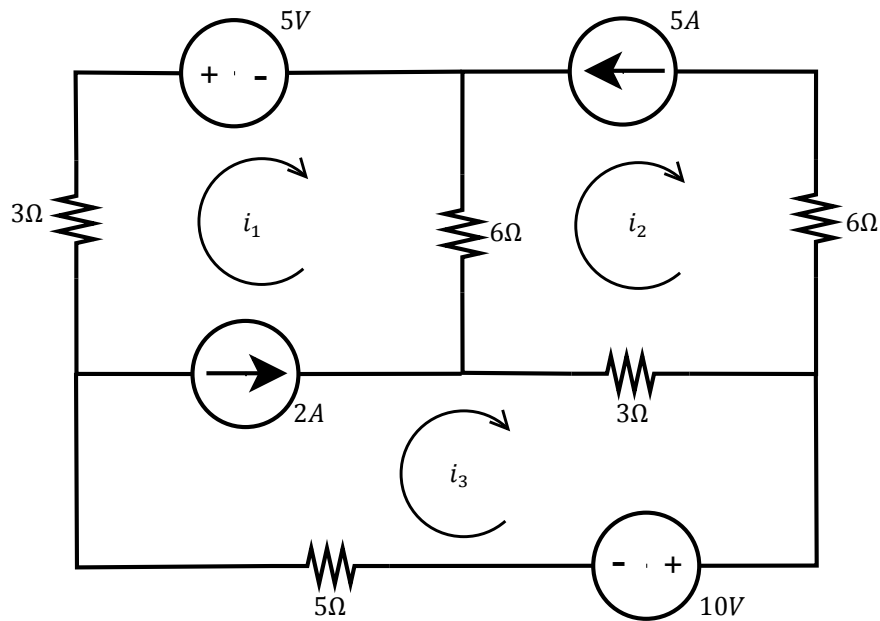
Gebruik lusanalyse om twee vergelykings van die volgende vorm op te stel:

$$ei_1 + fi_3 = 2 \text{ A}$$
$$gi_1 + hi_3 = -60 \text{ V (KVL)}$$

Question 2.2 [4]

Use mesh analysis to set up two equations of the following form:

$$ei_1 + fi_3 = 2 \text{ A}$$
$$gi_1 + hi_3 = -60 \text{ V (KVL)}$$



Vraag 2.3 [4]a) Bepaal V_2 .

$$\begin{bmatrix} 4 & -1 \\ -3 & 2 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 0 \\ -5 \end{bmatrix}$$

b) Bepaal i_2 .

$$\begin{bmatrix} 2 & -3 & 0 \\ -3 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

Question 2.3 [4]c) Determine V_2 .

$$\begin{bmatrix} 4 & -1 \\ -3 & 2 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 0 \\ -5 \end{bmatrix}$$

d) Determine i_2 .

$$\begin{bmatrix} 2 & -3 & 0 \\ -3 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

Vraag 2.4 [4]

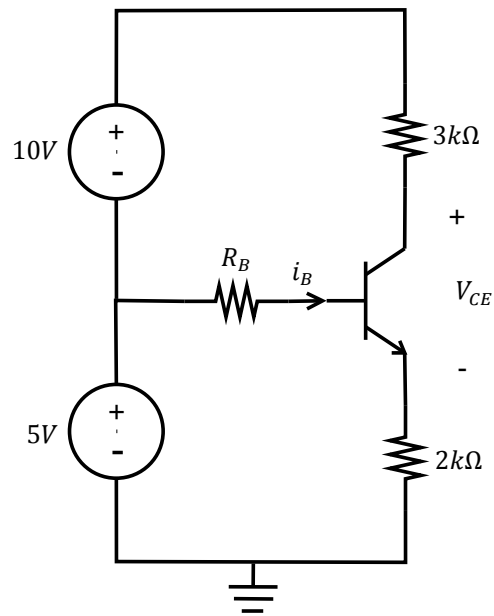
Gegee: $\beta = 100$; $V_{BE} = 0.7 \text{ V}$.

- a) Bepaal I_B as $R_B = 30 \text{ k}\Omega$.
b) Bepaal V_{CE} as $I_B = 20 \text{ }\mu\text{A}$.

Question 2.4 [4]

Given: $\beta = 100$; $V_{BE} = 0.7 \text{ V}$.

- a) Determine I_B if $R_B = 30 \text{ k}\Omega$.
b) Determine V_{CE} if $I_B = 20 \text{ }\mu\text{A}$.



Vraag 2.5 [4]

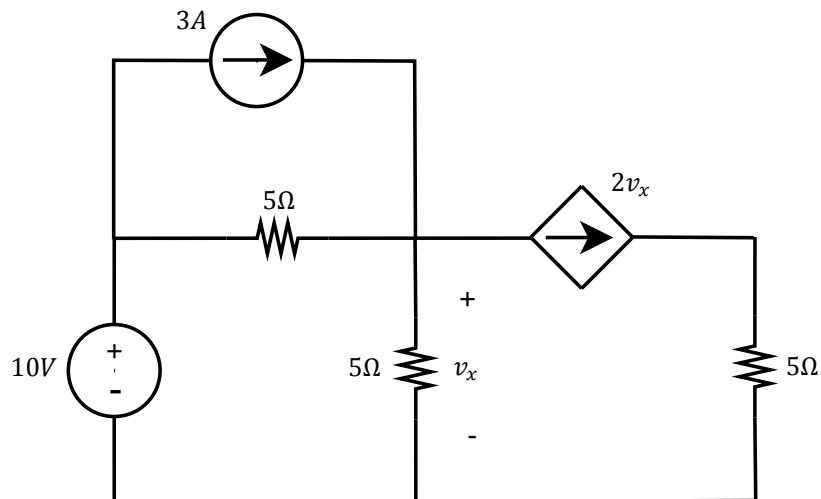
Gebruik die beginsel van superposisie:

- a) Bepaal v_x agv die 10V spanningsbron.
- b) Bepaal v_x agv die 3A stroombron.

Question 2.5 [4]

Use the concept of superposition:

- a) Determine v_x due to the 10V voltage source.
- b) Determine v_x due to the 3A current source.

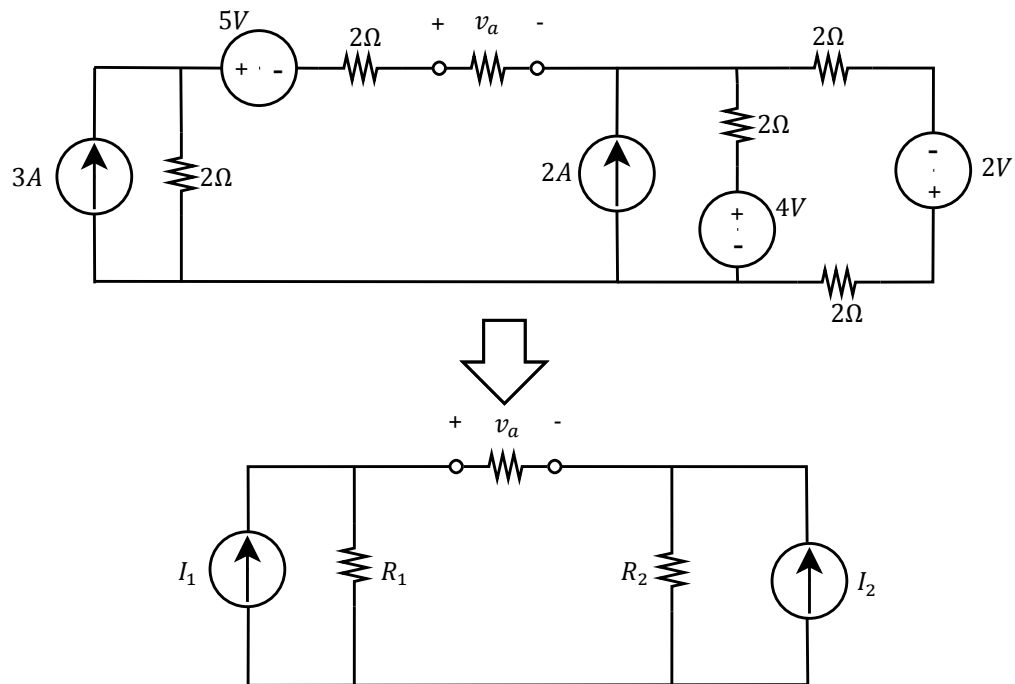


Vraag 2.6 [4]

Maak gebruik van brontransformatie en bepaal I_1 , R_1 , I_2 en R_2 .

Question 2.6 [4]

Use source transformation to determine I_1 , R_1 , I_2 and R_2 .

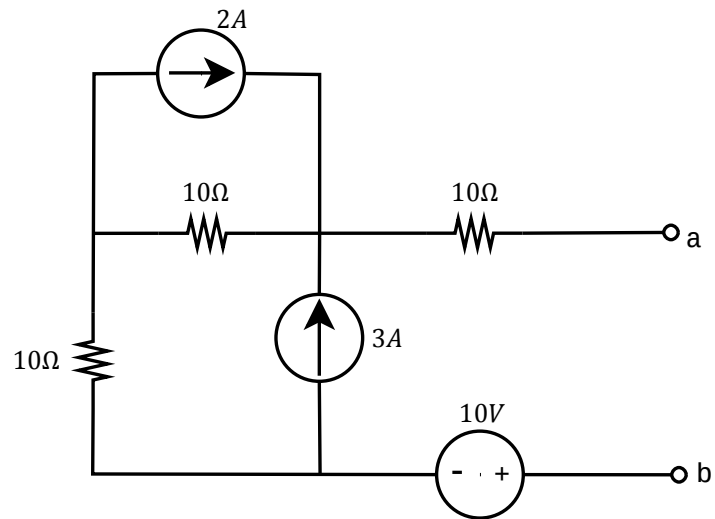


Vraag 2.7 [4]

Bepaal die Thevenin-ekwivalente baan ten opsigte van terminale a - b .

Question 2.7 [4]

Determine the Thevenin equivalent circuit with respect to terminals a - b .

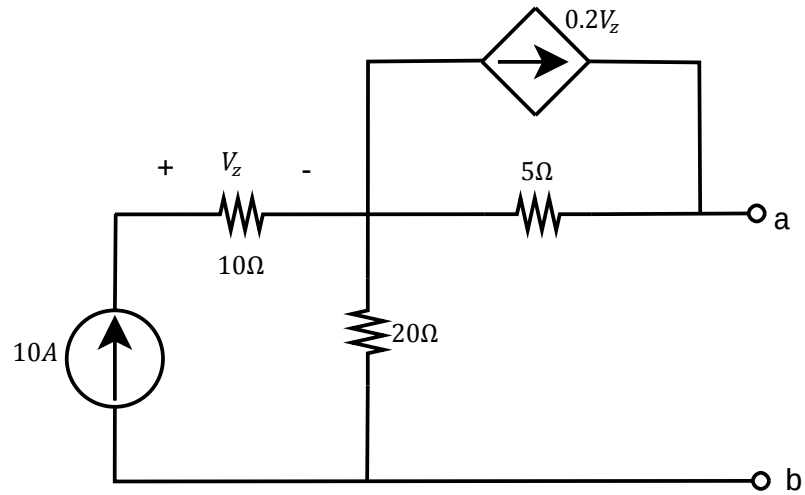


Vraag 2.8 [4]

Bepaal die Norton-ekwivalente baan ten opsigte van terminale a - b .

Question 2.8 [4]

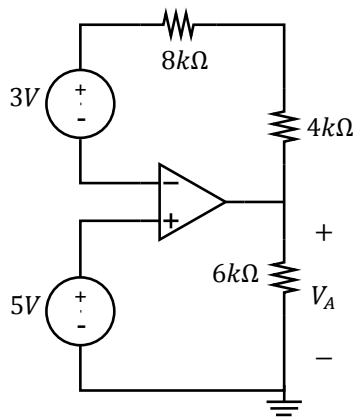
Determine the Norton equivalent circuit with respect to terminals a - b .



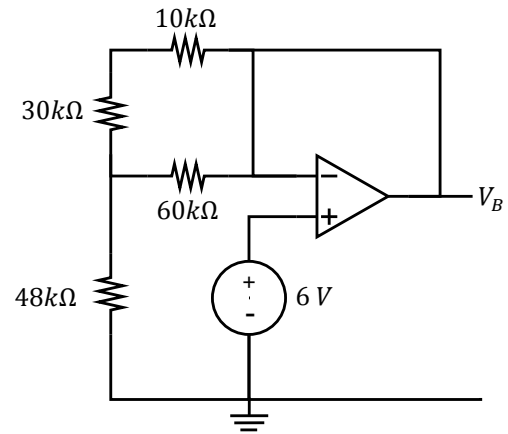
VRAAG/QUESTION 3 [18]

Vraag 3.1 [4]

Bepaal V_A en V_B vir kringbane 1 en 2 onderskeidelik.

Kringbaan / Circuit 1**Question 3.1 [4]**

Determine V_A and V_B for circuits 1 and 2 respectively.

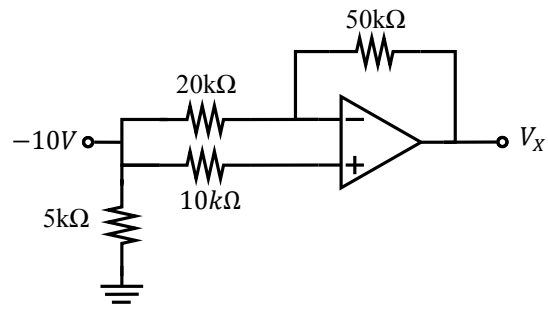
Kringbaan / Circuit 2

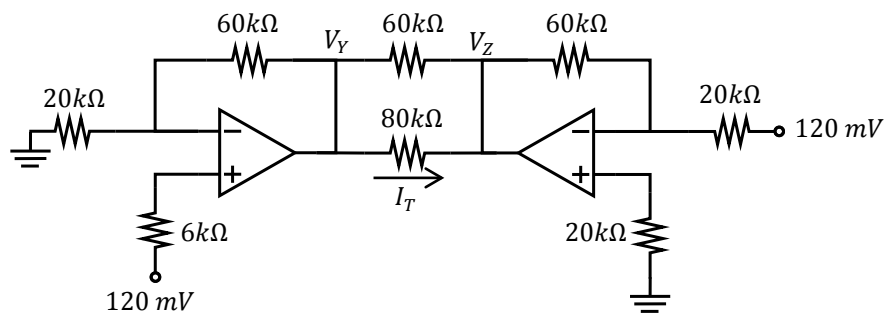
Vraag 3.2 [2]

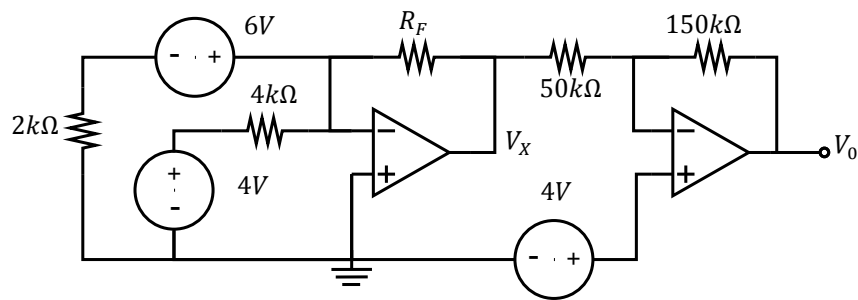
Bepaal V_X .

Question 3.2 [2]

Determine V_X .



Vraag 3.3 [3]Bepaal V_Y , V_Z en I_T .**Question 3.3 [3]**Determine V_Y , V_Z and I_T .

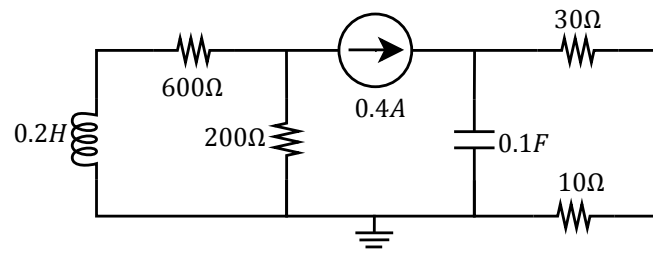
Vraag 3.4 [3]Indien $V_X = -10\text{ V}$, bepaal R_F , V_0 .**Question 3.4 [3]**If $V_X = -10\text{ V}$, determine R_F , V_0 .

Vraag 3.5 [2]

Bepaal die energie gestoor in die kapasitor en induktor.

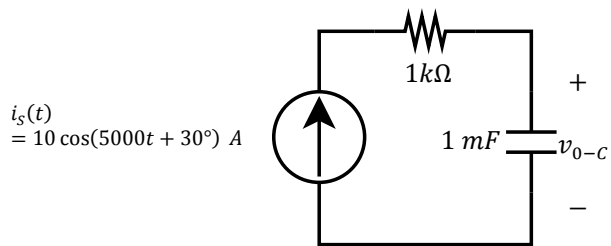
Question 3.5 [2]

Determine the energy stored in the capacitor and inductor.

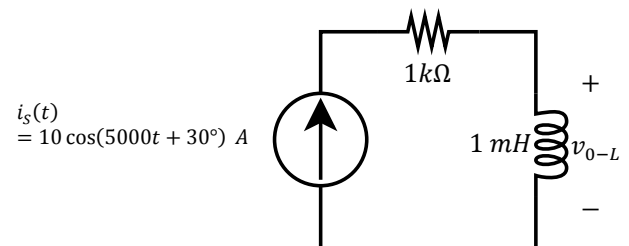


Vraag 3.6 [4]

Beskou kringbane 1 en 2. In altwee gevalle word die uitsetspanning gemeet oor die reaktiewe komponent in die kringbaan. Bepaal telkens v_{0-C} en v_{0-L} .

Kringbaan / Circuit 1**Question 3.6 [4]**

Consider circuits 1 and 2. For both circuits the output voltage is measured across the reactive component in the circuit. Determine v_{0-C} and v_{0-L} .

Kringbaan / Circuit 2

VRAAG/QUESTION 4 [22]

Vraag 4.1 [4]

a) Skryf $i(t)$ as 'n fasor in polêre vorm:

$$i(t) = 3 \cos(\omega t - 120^\circ) - 4 \sin(\omega t + 30^\circ) \text{ A}$$

b) Watter van die volgende seine loop voor en met hoeveel grade:

$$\bar{V}_1 = 5 \angle -70^\circ, v_2(t) = 10 \sin(\omega t + 100^\circ)$$

Question 4.1 [4]

a) Write $i(t)$ as a phasor in polar form:

$$i(t) = 3 \cos(\omega t - 120^\circ) - 4 \sin(\omega t + 30^\circ) \text{ A}$$

b) Which of the following signals is leading and by how many degrees:

$$\bar{V}_1 = 5 \angle -70^\circ, v_2(t) = 10 \sin(\omega t + 100^\circ)$$

Vraag 4.2 [4]a) Vereenvoudig en skryf die antwoord in **polêre vorm**:

$$\bar{I}_1 = \left(\frac{3\angle 60^\circ \times (6 + j8)}{-18 - j24} \right)^*$$

b) Vereenvoudig en skryf antwoord in **polêre vorm**:

$$\bar{V}_1 = \frac{\sqrt{2}e^{j45^\circ}}{\sqrt{8 + 8j}}$$

Question 4.2 [4]a) Simplify and write the answer in **polar form**:

$$\bar{I}_1 = \left(\frac{3\angle 60^\circ \times (6 + j8)}{-18 - j24} \right)^*$$

b) Simplify and write the answer in **polar form**:

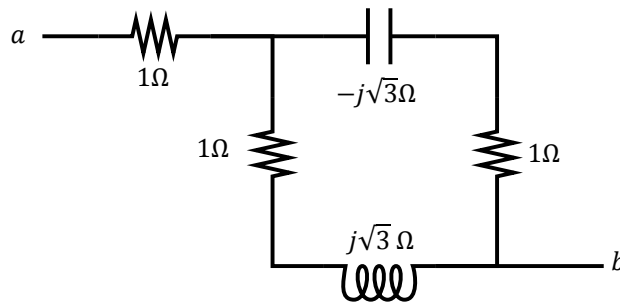
$$\bar{V}_1 = \frac{\sqrt{2}e^{j45^\circ}}{\sqrt{8 + 8j}}$$

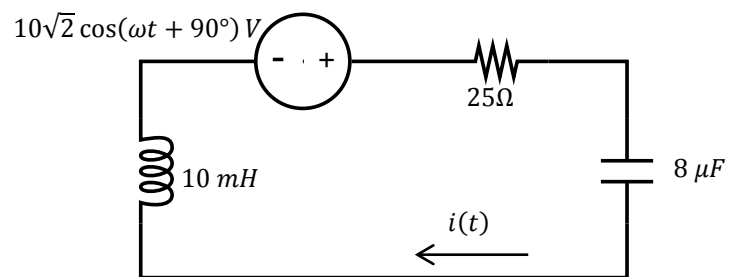
Vraag 4.3 [2]

Bepaal die totale impedansie Z_{ab} tussen terminale a en b .
Skryf die antwoord in reghoekige vorm.

Question 4.3 [2]

Determine the total impedance Z_{ab} between terminals a and b . Write the answer in rectangular form.



Vraag 4.4 [2]Bereken $i(t)$. ($\omega = 5 \text{ krad/s}$)**Question 4.4 [2]**Determine $i(t)$. ($\omega = 5 \text{ krad/s}$)

Vraag 4.5 [4]

Gebruik lus-analise om twee vergelykings van die volgende vorm op te stel:

$$(a_1 + jb_1)\bar{I}_1 + (a_2 + jb_2)\bar{I}_2 = -6j$$

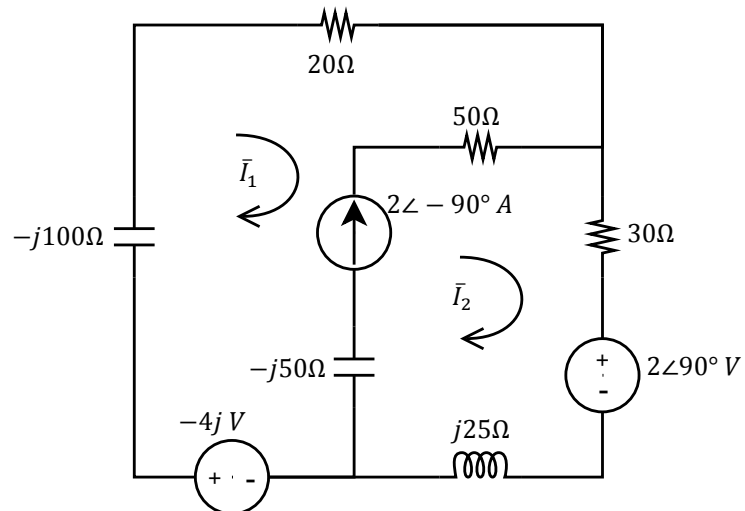
$$(a_3 + jb_3)\bar{I}_1 + (a_4 + jb_4)\bar{I}_2 = -2j$$

Question 4.5 [4]

Use mesh analysis to set up two equations of the following form:

$$(a_1 + jb_1)\bar{I}_1 + (a_2 + jb_2)\bar{I}_2 = -6j$$

$$(a_3 + jb_3)\bar{I}_1 + (a_4 + jb_4)\bar{I}_2 = -2j$$

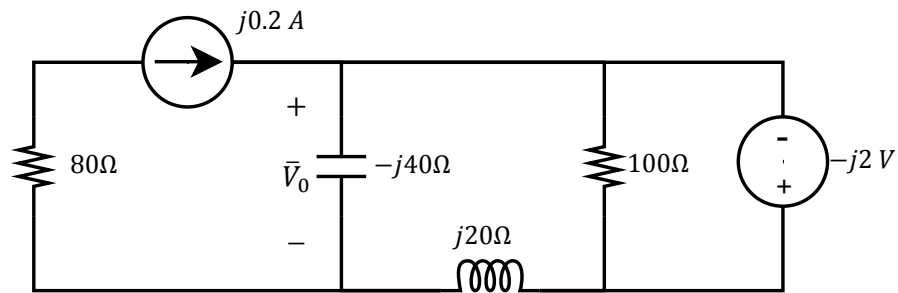


Vraag 4.6 [2]

Gebruik superpositie en bereken $\bar{V}_0$ slegs as gevolg van die stroombron.

Question 4.6 [2]

Use superposition and determine $\bar{V}_0$ as a result of the current source only.



Vraag 4.7 [4]

Bepaal die Thevenin-ekwivalent t.o.v. terminale a - b .
Skryf die antwoorde as fasors in reghoekige vorm.

Question 4.7 [4]

Find the Thevenin equivalent w.r.t. terminals a - b .
Write the answers as phasors in rectangular form.

